



COMPUTER NETWORKS

Computer Networks and the Internet

Team Networks

Department of Computer Science and Engineering

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Unit – 1 Computer Networks and the Internet

1.1 Introduction to Computer Networks

1.2 What is the Internet?

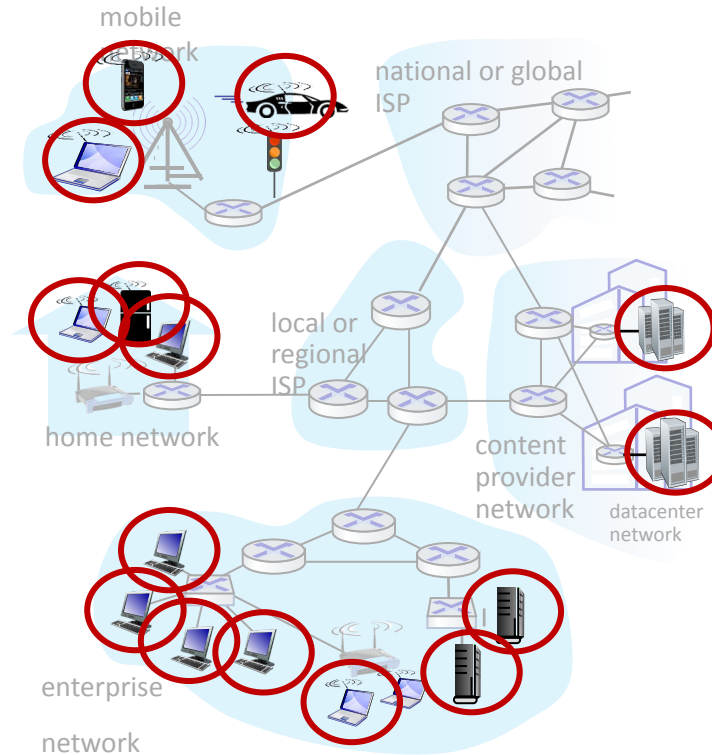
- A nuts-and-bolts and Services description, Protocol

1.3 Network edge

- End systems, Access networks, Physical media

Network edge:

- Hosts: clients & servers
- Servers in data centers



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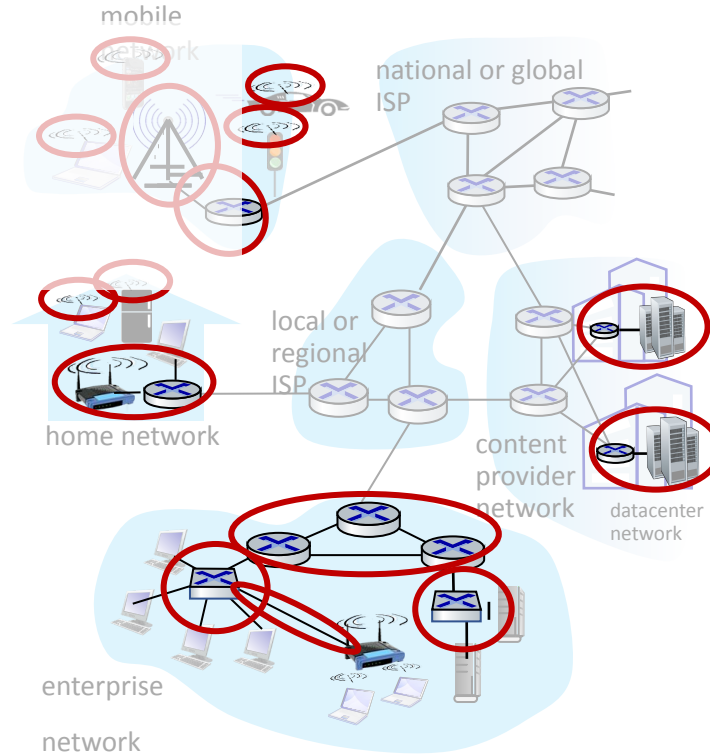
Network Edge: A closer look at network structure

Network edge:

- Hosts: clients & servers
- Servers in data centers

Access networks, physical media:

- wired, wireless communication links



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Network Edge: A closer look at network structure

Network edge:

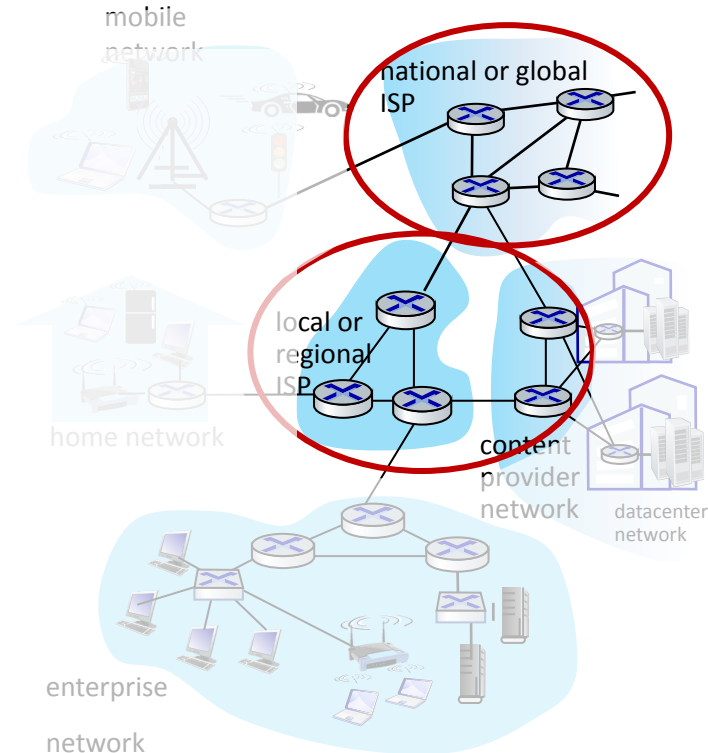
- Hosts: clients and servers
- Servers in data centers

Access networks, physical media:

- wired, wireless communication links

Network core:

- interconnected routers
- network of networks



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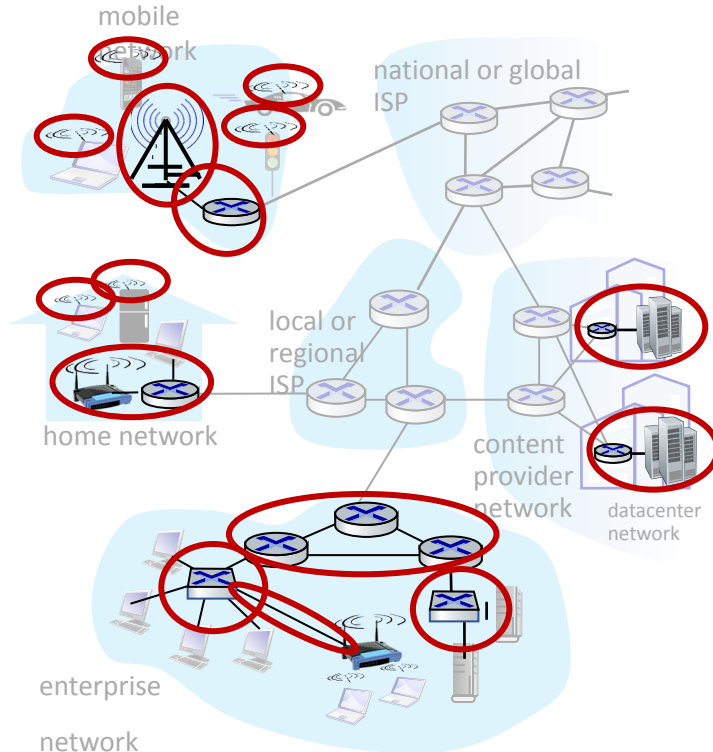
Network Edge: Access networks and Physical media

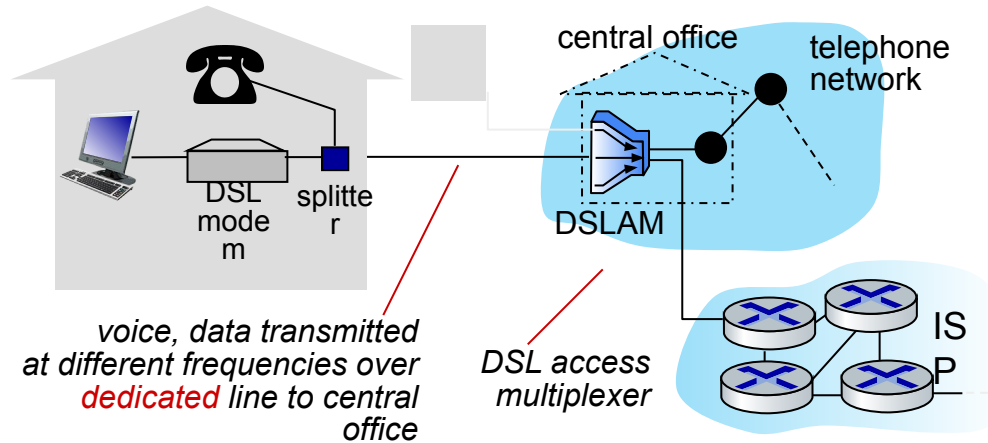
Q: How to connect end systems to edge router?

- Residential access networks
- Institutional access networks (school, company)
- Mobile access networks (WiFi, 4G/5G)

What to look for:

- Transmission rate (bits per second) of access network?
- Shared or dedicated access among users?



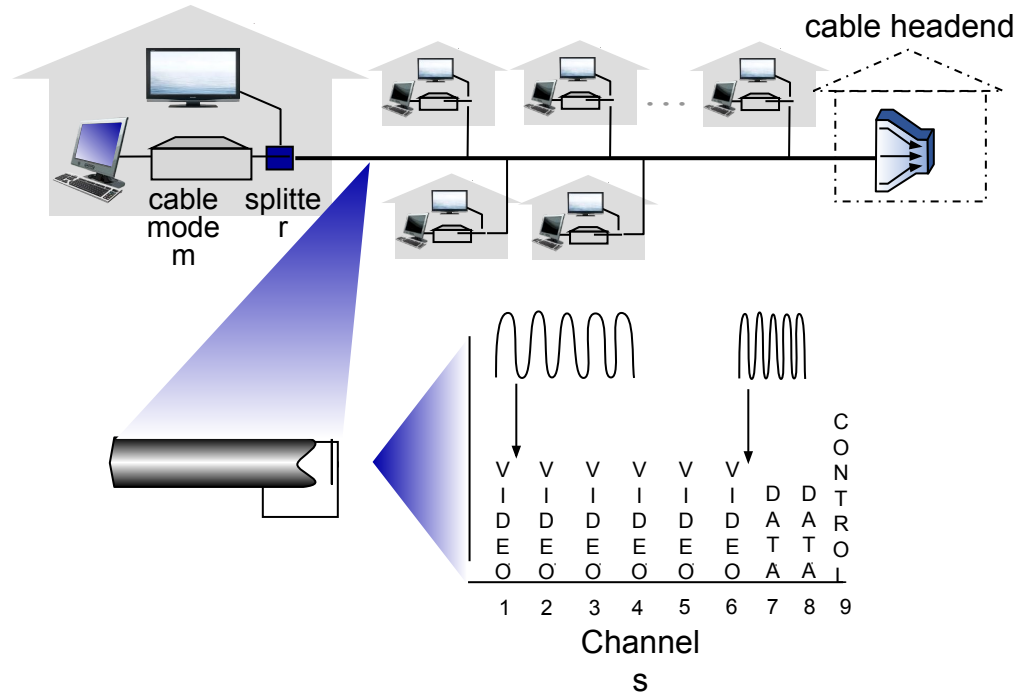


- 24-52 Mbps – downstream transmission rate
- 3.5-16 Mbps – upstream transmission rate
- Asymmetric access

- use **existing** telephone line to central office DSLAM
 - **data** over DSL phone line goes to Internet
 - **voice** over DSL phone line goes to telephone net
- A high-speed downstream channel, in the 50 kHz to 1 MHz band
- A medium-speed upstream channel, in the 4 kHz to 50 kHz band
- An ordinary two-way telephone channel, in the 0 to 4 kHz band

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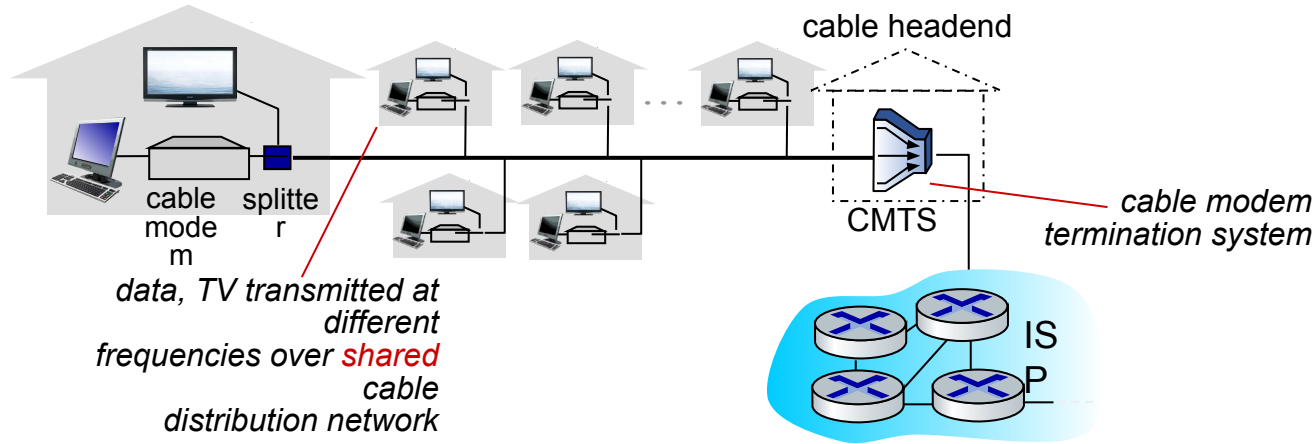
Network Edge: Access Networks: Cable-based access



Frequency division multiplexing (FDM): different channels transmitted in different frequency bands

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Network Edge: Access Networks: Cable-based access



▪ HFC: hybrid fiber coax

- Asymmetric:

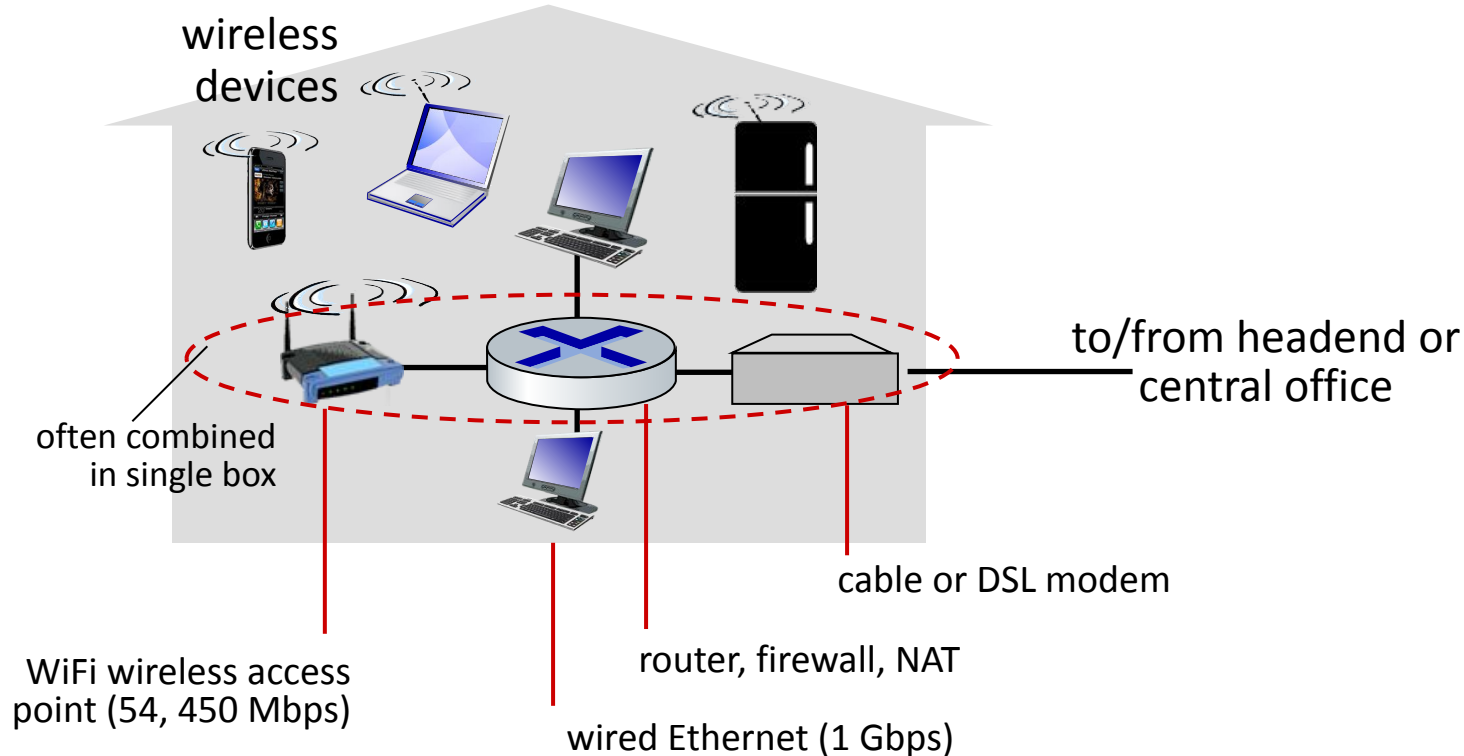
up to 40 Mbps – 1.2 Gbs downstream transmission rate,
30-100 Mbps upstream transmission rate

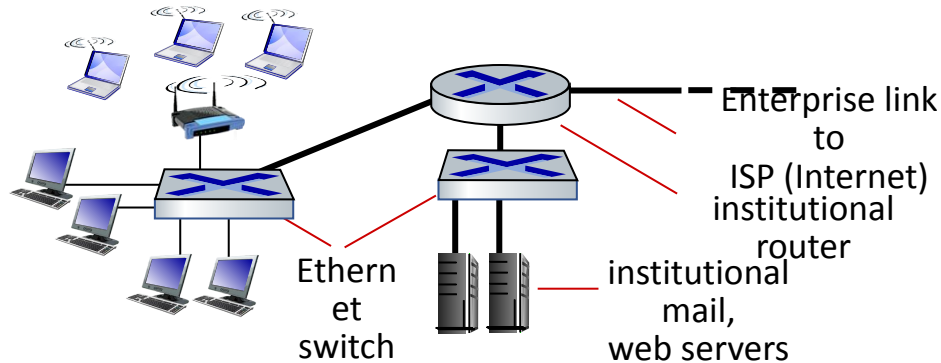
▪ Network of cable, fiber attaches homes to ISP router

- homes *share access network* to cable headend

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Network Edge: Access Networks – Home access





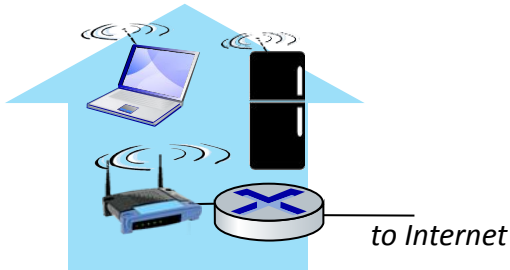
- companies, universities, etc.
- mix of wired, wireless link technologies, connecting a mix of switches and routers (we'll cover differences shortly)
 - Ethernet: wired access at 100Mbps, 1Gbps, 10Gbps
 - WiFi: wireless access points at 11, 54, 450 Mbps

Shared *wireless* access network connects end system to router

- via base station aka “access point”

Wireless local area networks (WLANs)

- typically within or around building (~100 ft)
- 802.11b/g/n (WiFi): 11, 54, 450 Mbps transmission rate



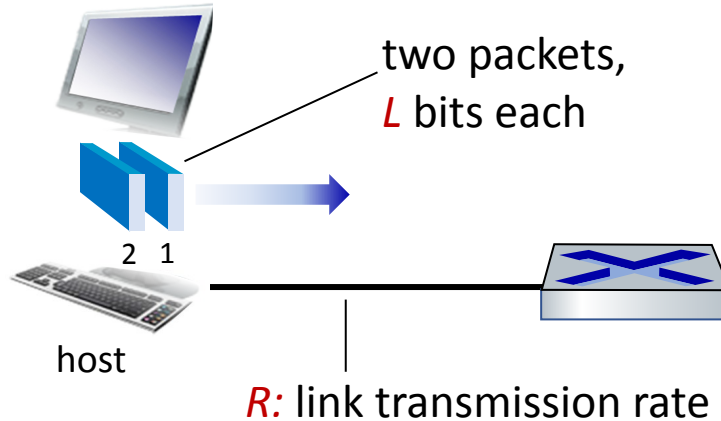
Wide-area cellular access networks

- provided by mobile, cellular network operator (10's km)
- 10's Mbps
- 4G cellular networks (5G coming)



Host sending function:

- takes application message
- breaks into smaller chunks, known as *packets*, of length L bits
- transmits packet into access network at *transmission rate R*
 - link transmission rate, aka link *capacity, aka link bandwidth*

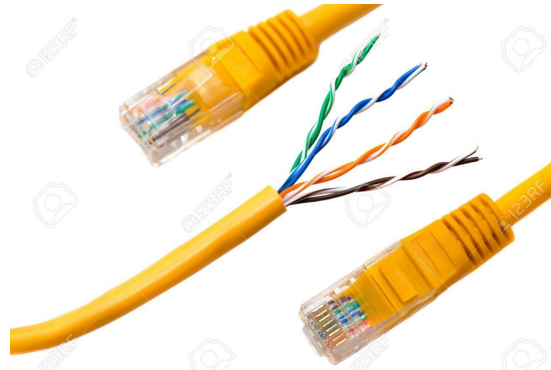


$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

- **bit:** propagates between transmitter/receiver pairs
- **physical link:** what lies between transmitter & receiver
- **guided media:**
 - signals propagate in solid media: copper, fiber, coax
- **unguided media:**
 - signals propagate freely, e.g., radio

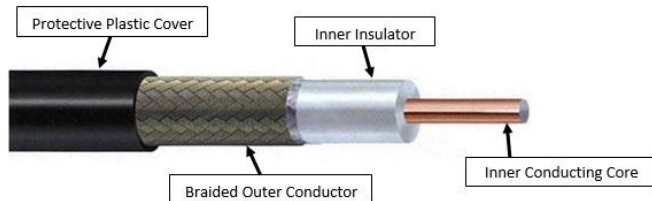
Twisted pair (TP)

- two insulated copper wires (STP & UTP)
 - Category 5: 100 Mbps, 1 Gbps Ethernet
 - Category 6: 10Gbps Ethernet



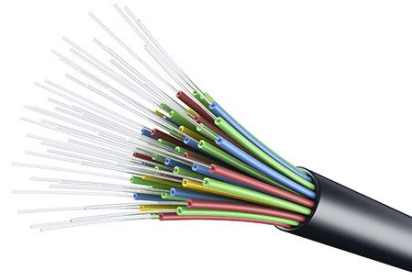
Coaxial cable:

- two concentric copper conductors
- concentric rather than parallel
- bidirectional
- broadband:
 - multiple frequency channels on cable
 - 100's Mbps per channel



Fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
 - high-speed point-to-point transmission (10's-100's Gbps)
- low error rate:
 - repeaters spaced far apart
 - immune to electromagnetic noise

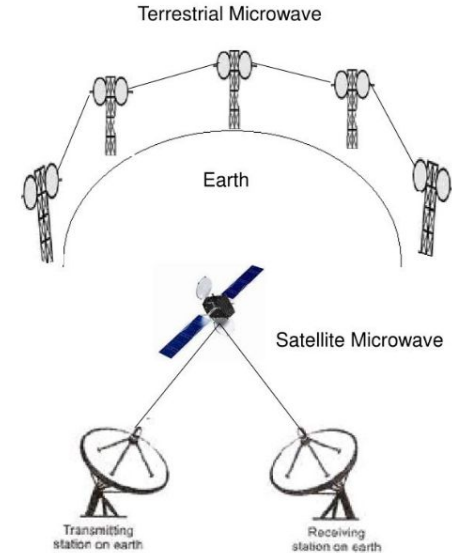


Wireless radio

- signal carried in electromagnetic spectrum
- no physical “wire”
- broadcast and “half-duplex” (sender to receiver)
- propagation environment effects:
 - reflection
 - obstruction by objects
 - interference

Radio link types:

- **terrestrial microwave**
 - up to 45 Mbps channels
- **Wireless LAN (WiFi)**
 - Up to 100's Mbps
- **wide-area** (e.g., cellular)
 - 4G cellular: ~ 10's Mbps
- **satellite**
 - up to 45 Mbps per channel
 - 280 msec end-end delay
 - geosynchronous vs. low-earth-orbit





THANK YOU

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